

THE FLAVOR FACTORY

Prepared for Church & Dwight / Therabreath

Strategic Flavor Partnership Booklet

Lakewood Implementation Support + Sensory Validation Plan

Compliance

documented
systems

Readiness

supply continuity

Validation

triangle testing

Innovation

oral-care flavor
design

Purpose of This Booklet

Position The Flavor Factory as a low-risk, high-capability oral-care partner.

The message we want Church & Dwight to feel:

Factory is organized, proactive, technically capable, and ready to s

1. Partnership

Frame TFF as a strategic partner, not a transactional vendor.

2. Confidence

Show C&D that implementation risk is being controlled.

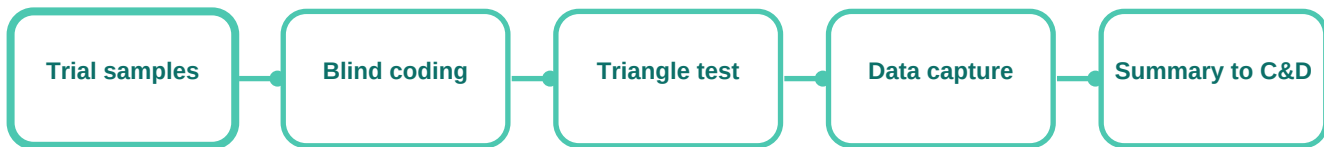
3. Action

Present sensory validation and data collection as a concrete next step.

Lakewood Mouthrinse Implementation Support

Proposed structure for product parity evaluation and external sensory screening.

- Align impacted variants and expected sample timing from Church & Dwight.
- Receive current approved control and Lakewood trial product for each variant.
- Blind code samples and run a structured triangle test with external participants.
- Collect forced-choice data, confidence level, sensory notes, preference, and comments.
- Summarize results by variant and provide Ross with a clean data package.



Proposed Variant Rollout

Phased plan based on trial timing and expected sample availability.

Variant	Trial Timing	Expected Evaluation Window	Priority
Invigorating Icy Mint	3rd week / end of June	Early to mid July	Phase 1
Mild Mint	End of June	Early to mid July	Phase 1
Rainforest Mint	End of June	Early to mid July	Phase 1
Whitening Dazzling Mint	Early July	Mid to late July	Phase 2
Overnight Chamomile Mint	Mid July	Late July / early Aug	Phase 2
Complete Revitalizing Mint	End of July	Early to mid Aug	Phase 3
Healthy Gums Clean Mint	Early Sept	Mid September	Phase 4
AntiCavity Sparkle Mint	Mid Sept	Late September	Phase 4



Triangle Testing Method

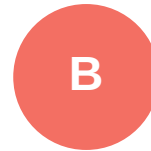
A forced-choice discrimination test: two samples are the same, one is different.



Code 101



Code 138



Code 175

Participant chooses the odd sample

Core Question

Can consumers detect a meaningful sensory difference between current and Lakewood trial product?

Why It Works

The method prevents bias because the participant does not know which sample is current or trial.

Output

A clean count of correct/incorrect selections plus confidence and sensory comments.

Participant Testing Protocol

Clean, repeatable process to keep the data credible.

- 1 Recruit** External participants; target 50-75 per variant where timing allows.
- 2 Screen** No illness, no food/drink/gum within 15 minutes, no known sensitivity to mouthrinse.
- 3 Code** Use randomized 3-digit sample codes; never use A/B labels on cups.
- 4 Taste** Taste in presented order; rinse with water between each sample.
- 5 Choose** Forced choice: select which sample is different, even if unsure.
- 6 Capture** Record confidence, sensory notes, optional preference, and open comments.

Sample Randomization Plan

Balanced serving orders reduce order bias and strengthen the result.

Presentation Order	Cup 1	Cup 2	Cup 3	Odd Sample
AAB	A	A	B	3
ABA	A	B	A	2
BAA	B	A	A	1
BBA	B	B	A	3
BAB	B	A	B	2
ABB	A	B	B	1

Coordinator note: rotate through these six orders evenly for each variant. Use random 3-digit codes on cups and record the key on the coordinator sheet only.

External Participant Plan

Chapman students + local external participants create a practical screening panel.

Target Count

Preferred: 50-75 participants per variant.
Minimum useful screen: 30 participants per variant.

Recruiting Source

Chapman University contact through Tasha / Zoe, plus local external participants if needed.

Incentive

Recommended: \$10-\$15 gift card per participant or raffle + refreshments for larger groups.

- Keep sessions short: target 8-12 minutes per participant per variant.
- Avoid telling participants which product is current or trial.
- Do not ask preference until after the forced-choice odd-sample question.
- Keep setup clean, simple, and consistent: water, spit cups, neutral crackers, coded cups, forms.

Triangle Test Form Fields

What each participant completes.

Field	Purpose
Participant ID	Keep responses anonymous and sortable.
Product Variant	Tie response to the correct Therabreath variant.
Sample Set Code	Connects participant to the randomized serving order.
Odd Sample Selection	Primary triangle test output.
Confidence Level	Adds interpretation context.
Difference Noted	Shows type of sensory difference detected.
Optional Preference	Secondary insight; not used for primary triangle result.
Comments	Captures useful language for Ross / C&D review.

Decision Framework

The triangle test is evaluated by correct odd-sample selections vs. chance guessing.

- Because each participant has a 1-in-3 chance of guessing correctly, the key output is the number of correct odd-sample selections.
- Results should be summarized by variant with total participants, correct selections, percent correct, confidence profile, and common sensory comments.
- Use the data as an external discrimination screen - not a clinical claim or national consumer study.

Panel Size	Chance Expected	Strong Signal if Correct >=
30	10	15-16+
50	16-17	24-25+
60	20	28-29+
75	25	34-35+

Note: final statistical interpretation should be reviewed before sending results externally. The workbook included with this packet calculates counts and percentages and can be expanded for formal significance testing.

Proposed Data Package for Ross

A clean handoff that makes decision-making easier.

1. Variant Summary

High-level table by variant: N, correct count, percent correct, confidence profile, and top sensory comments.

2. Raw Data File

An Excel workbook with participant-level rows, randomized order, selection, confidence, sensory notes, and comments.

3. Interpretation Notes

Plain-English conclusion: likely perceptible difference, no strong signal, or needs retest.

4. Next-Step Recommendation

If difference is detected, identify sensory direction: mint intensity, bitterness, burn/cooling, aftertaste, mouthfeel, aroma.

Goal: give C&D usable evidence, not just opinions.

Why This Elevates The Flavor Factory

This turns an implementation issue into a partnership-strengthening system.

- Shows proactive risk management around Lakewood implementation.
- Gives Church & Dwight a structured external feedback loop before broader rollout.
- Demonstrates sensory discipline and oral-care category understanding.
- Creates a repeatable process that can support future innovations, reformulations, and site transitions.

Supplier behavior says “we can make it.”

Partner behavior says “we will help you prove it works.”

Existing Documentation to Integrate

The uploaded packet contains the proof points that support the partnership story.



Formulation Guidelines



OU Kosher Certification



Halal Certificate



Emergency Agreement



Approved Materials / Locations



Product Specification Example

Use these as evidence, not clutter. In the final print version, the scanned documents should be placed in an appendix or redesigned as clean proof-point pages with source scans available upon request.

From Document Archive to Executive Story

Convert raw files into a clear trust-building narrative.

Formulation Alignment

Purpose-driven formulation, simple ingredient philosophy, and preference for natural, organic, and non-GMO where possible.

Certification Readiness

Kosher, Halal, food-safety, organic, and emergency manufacturing documentation tell a stronger compliance story.

Specification Control

Product specification sheets show traceability, QA discipline, shelf-life guidance, and controlled customer-specific documentation.

Supply Continuity

Guidelines, lead times, alternate manufacturing, and location coordination support business continuity messaging.

Triangle Test Coordinator Checklist

What needs to be ready before the first testing session.

- ☐ Confirm product variants and sample volumes.
- ☐ Define control vs. Lakewood trial sample key.
- ☐ Generate 3-digit random sample codes.
- ☐ Print participant forms and coordinator key sheets.
- ☐ Prepare coded cups, water, spit cups, neutral crackers, gloves, waste bags, pens.
- ☐ Train coordinator: no coaching, no preference discussion before forced-choice question.
- ☐ Collect forms immediately and enter data into workbook.
- ☐ Review results by variant before sending to Church & Dwight.

Chapman University Outreach Angle

Simple, credible language for recruiting external participants.

We are exploring a structured sensory discrimination exercise for oral-care products. Participants would be asked to taste three coded mouthrinse samples and identify which sample is different. The session is short, anonymous, and designed to collect aggregate sensory feedback. Participants would receive a small gift card or similar thank-you incentive.

What to Ask Chapman

Can a professor, department, or student group help coordinate external participants for a short sensory screening?

What Not to Say

Do not position this as medical testing, claims substantiation, or a formal clinical study.

Best Fit

Food science, sensory science, marketing research, consumer behavior, or entrepreneurship groups.

Immediate Next Actions

Move from concept to controlled execution.

Today

Finalize draft testing protocol and participant form.

This week

Confirm Chapman contact and available participant pool.

Before samples arrive Build randomization sheets, sample coding, and data workbook.

After first samples Run Phase 1 pilot with 10-12 participants to validate process.

Full test

Run target panel size and summarize results by variant.

C&D review

Send summary + raw data + recommendation to Ross for alignment.

Built for partnership. Designed for confidence.

The Flavor Factory can help Church & Dwight move quickly without flying blind.

Compliance

Operational readiness

Sensory validation

Oral-care innovation